

Tiling-count sequence of the $3 \times n$ rectangle with a
side-middle cell removed: interleave identity and
parity/mod-4 laws
(Lean 4 machine-checked claim of record)

Aurora0134*

September 28, 2026

Abstract

Let $a(n)$ count the domino tilings of a $3 \times n$ rectangle with the middle cell of its last column removed. The sequence $1, 8, 46, 295, 1814, 11306, \dots$ is defined by six seeds and the order-six sparse recurrence $a(n+6) = 4a(n+5) + 14a(n+4) - 10a(n+2) + a(n)$. We state and machine-check, in Lean 4 (v4.34.0, mathlib v4.34.0), three theorems: the main sequence is the interleaving of its odd- and even-indexed subsequences, which satisfy one and the same order-six recurrence; $a(n)$ is odd if and only if $n \equiv 1 \pmod{3}$; and $a(n) \bmod 4$ depends only on $n \bmod 12$ with pattern $(1, 0, 2, 3, 2, 2, 1, 2, 0, 3, 0, 0)$. Every proof compiles with no `sorry` and `#print axioms` reports only `{propext, Quot.sound}`. Novelty adjudication: no occupying record found in three review rounds; value tier *new sequence* — the recurrence’s order and coefficients are shared with A033506 and no novelty is claimed for them. The combinatorial identification is not kernel-proved and is stated as a conjecture. This note is a priority claim of record, not a full paper.

1 Introduction

The object of this note is the tiling-count sequence of the $3 \times n$ rectangle with a side-middle cell removed: tiled by 1×2 dominoes, the number of tilings is

*Contact: via the GitHub account Aurora0134; ORCID placeholder, to be filled in by the author at Zenodo upload. No personal information is carried in this note.

$a(n)$, and the sequence begins 1, 8, 46, 295, 1814, 11306, 70161, ... for $n \geq 1$ (the matching-polynomial setting [8]). Three pure-sequence theorems about it — an interleave identity and two modular laws — were produced by the AI4Math pipeline, in which LLM workers generate statements and proof scripts while the Lean 4 kernel is the sole adjudicator at every gate: statements are frozen by hash before proving, and a separate audit re-computes the symmetric difference, scans the whole file for **sorry/admit** including comments, and re-prints the axiom dependencies [1, 2]. Verification strength is stated below verbatim. This note is a priority claim of record deposited so that a dated, citable public record exists ahead of any narrative paper; the fuller paper for the same theorems is prepared separately and will reference this deposit. No literature review is attempted here.

2 Statement

Indexing conventions, locked on the source task: 0-based in the Lean layer, 1-based in the counting reading, through $A(n) = a(n+1)$, $O(k) = a(2k+1)$ and $E(k) = a(2k+2)$; the Lean sequences **asid**, **osid** and **esid** are A , O and E . The statements below are about sequences defined by literal initial values and integer-coefficient recurrences only; no graph object appears in the frozen statements. The main sequence is defined by six initial values and the order-six sparse recurrence

$$a(n+6) = 4a(n+5) + 14a(n+4) - 10a(n+2) + a(n),$$

whose signature $(4, 14, 0, -10, 0, 1)$ is the one registered for the undefected $3 \times n$ grid, A033506 [11]; nothing below depends on, and nothing below claims anything about, that sharing.

Listing 1: The main sequence **asid**, verbatim from the frozen statement file `formalized/01-comb08-sidemid-statements.lean` (lines 28–39, sha256 5fe889d5).

```

1  /-- **序列 asid (主列, 六阶稀疏递推) **.
2  初值  $a(0..5) = 1, 8, 46, 295, 1814, 11306$ ;
3  递推  $a(n+6) = 4 \cdot a(n+5) + 14 \cdot a(n+4) - 10 \cdot a(n+2) + a(n)$  ( $n \geq 0$ ;
4  递推签名  $-(4, 14, 0, 10, 0, 1)$ , 系数为 0 的滞后项按主列 def 口径省略不
   写)。 -/
5  def asid : ℕ → ℤ
6    | 0 => 1
7    | 1 => 8
8    | 2 => 46

```

```

9 | 3 => 295
10 | 4 => 1814
11 | 5 => 11306
12 | n + 6 => 4 * asid (n + 5) + 14 * asid (n + 4) - 10 * asid (n +
    2) + asid n

```

Theorem 1 (interleaving). *For every $k \in \mathbb{N}$,*

$$A(2k) = O(k) \quad \text{and} \quad A(2k+1) = E(k).$$

In the 1-based reading this is $a(2k+1) = o(k)$ and $a(2k+2) = e(k)$: the main sequence is the interleaving of its odd-indexed and even-indexed subsequences, and those two subsequences satisfy one and the same order-six recurrence.

Listing 2: Sequences `osid`, `esid` and the statement of Theorem 1, verbatim from the frozen statement file (lines 41–74, sha256 5fe889d5). The `:=` by at the end of the theorem line opens the proof body, which is given in Listing 6; the `sorry` placeholder of the frozen file is not part of the excerpt.

```

1  /-- **序列 osid (1-based 奇位子列, 六阶递推) **.
2  初值 o(0..5) = 1, 46, 1814, 70161, 2708104, 104507480
3  (= asid 在下标 0,2,4,6,8,10 处之值, 即 a(1),a(3),a(5),a(7),a(9),a
   (11));
4  递推 o(k+6) = 44·o(k+5) - 216·o(k+4) + 282·o(k+3) - 128·o(k+2) + 20
   ·o(k+1) - o(k) (k ≥ 0)。 -/
5  def osid : ℕ → ℤ
6    | 0 => 1
7    | 1 => 46
8    | 2 => 1814
9    | 3 => 70161
10   | 4 => 2708104
11   | 5 => 104507480
12   | k + 6 =>
13     44 * osid (k + 5) - 216 * osid (k + 4) + 282 * osid (k + 3)
14     - 128 * osid (k + 2) + 20 * osid (k + 1) - osid k
15
16 /-- **序列 esid (1-based 偶位子列, 六阶递推) **.
17 初值 e(0..5) = 8, 295, 11306, 435986, 16823455, 649209736
18 (= asid 在下标 1,3,5,7,9,11 处之值, 即 a(2),a(4),a(6),a(8),a(10),a
   (12));
19 递推 e(k+6) = 44·e(k+5) - 216·e(k+4) + 282·e(k+3) - 128·e(k+2) + 20
   ·e(k+1) - e(k) (k ≥ 0)。 -/
20 def esid : ℕ → ℤ
21   | 0 => 8

```

```

22 | 1 => 295
23 | 2 => 11306
24 | 3 => 435986
25 | 4 => 16823455
26 | 5 => 649209736
27 | k + 6 =>
28   44 * esid (k + 5) - 216 * esid (k + 4) + 282 * esid (k + 3)
29   - 128 * esid (k + 2) + 20 * esid (k + 1) - esid k
30
31 /-- **T1 (主攻) **: 主列六阶递推 = 两条同系数六阶子列的交织——
32 对一切 k, asid (2k) = osid k 且 asid (2k+1) = esid k。 -/
33 theorem asid_interleave (k : ℕ) :
34   asid (2 * k) = osid k ∧ asid (2 * k + 1) = esid k := by

```

Theorem 2 (modulo 2 pattern). *For every $n \in \mathbb{N}$ and every r with $n \bmod 3 = r$,*

$$A(n) \bmod 2 = P_3(r), \quad P_3(0) = 1, \quad P_3(r) = 0 \text{ for } r \neq 0.$$

In the 1-based convention: $a(n)$ is odd if and only if $n \equiv 1 \pmod{3}$.

Listing 3: Pattern `pat3` and the statement of Theorem 2, verbatim from the frozen statement file (lines 77–86, sha256 5fe889d5).

```

1 /-- **余数样式 pat3**: 0↦1, 其余↦0
2 (mod 2 余数 0-based 口径: asid n 为奇 ⇔ n ≡ 0 (mod 3))。 -/
3 def pat3 : ℕ → ℤ
4   | 0 => 1
5   | _ => 0
6
7 /-- **T2 (伴随) **: asid 的 mod 2 余数仅依赖 n mod 3, 样式为 pat3
8 (即 asid n 为奇 ⇔ n ≡ 0 (mod 3); 1-based 口径 = a(n) 为奇 ⇔
9   n ≡ 1 (mod 3))。 -/
10 theorem asid_mod2_period3 (n r : ℕ) (hr : n % 3 = r) :
    asid n % 2 = pat3 r := by

```

Theorem 3 (modulo 4 pattern). *For every $n \in \mathbb{N}$ and every r with $n \bmod 12 = r$,*

$$A(n) \bmod 4 = P_{12}(r),$$

where P_{12} on the residues $0, \dots, 11$ is $(1, 0, 2, 3, 2, 2, 1, 2, 0, 3, 0, 0)$. In the 1-based convention: $a(n) \bmod 4$ depends only on $n \bmod 12$, with the same pattern along the residues $1, \dots, 12$. The theorem implies Theorem 2: the odd values occur exactly at the residues $0, 3, 6, 9$ modulo 12, that is, at $n \equiv 0 \pmod{3}$ in the 0-based reading.

Listing 4: Pattern `pat12` and the statement of Theorem 3, verbatim from the frozen statement file (lines 89–108, sha256 5fe889d5).

```

1  /-- **余数样式 pat12** (12 留数类  $\mapsto$  mod 4 余数) :
2  0 $\mapsto$ 1, 1 $\mapsto$ 0, 2 $\mapsto$ 2, 3 $\mapsto$ 3, 4 $\mapsto$ 2, 5 $\mapsto$ 2, 6 $\mapsto$ 1, 7 $\mapsto$ 2, 8 $\mapsto$ 0, 9 $\mapsto$ 3, 10 $\mapsto$ 0,
   11 $\mapsto$ 0. -/
3  def pat12 :  $\mathbb{N} \rightarrow \mathbb{Z}$ 
4      | 0 => 1
5      | 1 => 0
6      | 2 => 2
7      | 3 => 3
8      | 4 => 2
9      | 5 => 2
10     | 6 => 1
11     | 7 => 2
12     | 8 => 0
13     | 9 => 3
14     | 10 => 0
15     | _ => 0
16
17  /-- **T3 (伴随) **: asid 的 mod 4 余数仅依赖  $n \bmod 12$ , 样式为 pat12
18  (蕴含 T2: 奇值恰在  $n \% 12 \in \{0, 3, 6, 9\} \iff n \equiv 0 \pmod{3}$ ) 。 -/
19  theorem asid_mod4_period12 (n r :  $\mathbb{N}$ ) (hr : n % 12 = r) :
20      asid n % 4 = pat12 r := by

```

Conjecture 1 (combinatorial bridge, not proved here). For every $n \geq 1$, $A(n-1) = a(n)$, where a counts the matchings (including the empty one) of the graph obtained from the $3 \times n$ grid by deleting the middle vertex of the last column.

3 Proof

All proofs share one skeleton: strong induction on the index (`Nat.strong_induction_on`), the seed window discharged as literal cases by `interval_cases` plus `decide`, one definitional unfolding of the recurrence in the inductive step (`simp only [asid]` followed by `omega`), and residue-class case analysis by `mod_cases` where the goal branches on a residue. The auxiliary twelve-step even-lag identity `asid_evenlag12` (Listing 5) is what makes the term-by-term matching in the interleave proof possible; its complete proof, together with the one-step unfolding `asid_shift` it builds on, is reproduced verbatim below. The complete proofs of Theorems 1 and 2 follow verbatim; for Theorem 3 the statement, the induction skeleton and

the first of the twelve `mod_cases` branches are excerpted — the remaining eleven branches are identical in shape (lines 183–237 of the final file) and the complete 238-line file ships in this deposit under `proofs/`.

Listing 5: The auxiliary lemmas `asid_shift` and `asid_evenlag12` with complete proofs, verbatim from `final/01-comb08-sidemid-proved.lean` (lines 80–115, sha256 19b5c390). These two declarations are proof-body scaffolding added after the freeze and therefore do not occur in the frozen statement file.

```

1  /-- 辅助: asid 单步展开 (t+6 形)。 -/
2  theorem asid_shift (t : ℕ) :
3      asid (t + 6) = 4 * asid (t + 5) + 14 * asid (t + 4) - 10 *
4      asid (t + 2) + asid t := by
5      simp only [asid]
6
7  /-- 辅助: 12 阶偶滞后恒等式  $asid(m+12) = 44 \cdot asid(m+10) - 216 \cdot asid(m+8) + 282 \cdot asid(m+6) - 128 \cdot asid(m+4) + 20 \cdot asid(m+2) - asid(m)$ 。全级联展开到基  $asid(m..m+5)$  后 ring 闭合。 -/
8  theorem asid_evenlag12 (m : ℕ) :
9      asid (m + 12) = 44 * asid (m + 10) - 216 * asid (m + 8) + 282
10     * asid (m + 6)
11     - 128 * asid (m + 4) + 20 * asid (m + 2) - asid m := by
12     rw [show m + 12 = (m + 6) + 6 by ring, asid_shift (m + 6),
13         show m + 6 + 5 = m + 11 by ring,
14         show m + 6 + 4 = m + 10 by ring,
15         show m + 6 + 2 = m + 8 by ring]
16     rw [show m + 11 = (m + 5) + 6 by ring, asid_shift (m + 5),
17         show m + 5 + 5 = m + 10 by ring,
18         show m + 5 + 4 = m + 9 by ring,
19         show m + 5 + 2 = m + 7 by ring]
20     rw [show m + 10 = (m + 4) + 6 by ring, asid_shift (m + 4),
21         show m + 4 + 5 = m + 9 by ring,
22         show m + 4 + 4 = m + 8 by ring,
23         show m + 4 + 2 = m + 6 by ring]
24     rw [show m + 9 = (m + 3) + 6 by ring, asid_shift (m + 3),
25         show m + 3 + 5 = m + 8 by ring,
26         show m + 3 + 4 = m + 7 by ring,
27         show m + 3 + 2 = m + 5 by ring]
28     rw [show m + 8 = (m + 2) + 6 by ring, asid_shift (m + 2),
29         show m + 2 + 5 = m + 7 by ring,
30         show m + 2 + 4 = m + 6 by ring,
31         show m + 2 + 2 = m + 4 by ring]
32     rw [show m + 7 = (m + 1) + 6 by ring, asid_shift (m + 1),

```

```

32     show m + 1 + 5 = m + 6 by ring,
33     show m + 1 + 4 = m + 5 by ring,
34     show m + 1 + 2 = m + 3 by ring]
35   rw [asid_shift m]
36   ring

```

Listing 6: Complete proof of Theorem 1, verbatim from final /01-comb08-sidemid-proved.lean (lines 117–142, sha256 19b5c390).

```

1  /-- **T1 (主攻) **: 主列六阶递推 = 两条同系数六阶子列的交织——
2  对一切  $k$ ,  $asid(2k) = osid\ k$  且  $asid(2k+1) = esid\ k$ . -/
3  theorem asid_interleave (k : ℕ) :
4    asid (2 * k) = osid k ∧ asid (2 * k + 1) = esid k := by
5    induction k using Nat.strong_induction_on with
6    | h k ih =>
7      by_cases hk : k < 6
8      · interval_cases k <|> decide
9      · obtain ⟨j, rfl⟩ : ∃ j, k = j + 6 := ⟨k - 6, by omega⟩
10       constructor
11       · rw [show 2 * (j + 6) = (2 * j) + 12 by ring,
12         asid_evenlag12 (2 * j)]
13         simp only [osid]
14         rw [show 2 * j + 10 = 2 * (j + 5) by ring, (ih (j + 5) (by
15           omega)).1,
16           show 2 * j + 8 = 2 * (j + 4) by ring, (ih (j + 4) (by
17             omega)).1,
18           show 2 * j + 6 = 2 * (j + 3) by ring, (ih (j + 3) (by
19             omega)).1,
20           show 2 * j + 4 = 2 * (j + 2) by ring, (ih (j + 2) (by
21             omega)).1,
22           show 2 * j + 2 = 2 * (j + 1) by ring, (ih (j + 1) (by
23             omega)).1,
24           (ih j (by omega)).1]
25         · rw [show 2 * (j + 6) + 1 = (2 * j + 1) + 12 by ring,
26           asid_evenlag12 (2 * j + 1)]
27           simp only [esid]
28           rw [show 2 * j + 1 + 10 = 2 * (j + 5) + 1 by ring, (ih (j
29             + 5) (by omega)).2,
30             show 2 * j + 1 + 8 = 2 * (j + 4) + 1 by ring, (ih (j +
31               4) (by omega)).2,
32             show 2 * j + 1 + 6 = 2 * (j + 3) + 1 by ring, (ih (j +
33               3) (by omega)).2,
34             show 2 * j + 1 + 4 = 2 * (j + 2) + 1 by ring, (ih (j +
35               2) (by omega)).2,

```

```

25      show 2 * j + 1 + 2 = 2 * (j + 1) + 1 by ring, (ih (j +
26      1) (by omega)).2,
      (ih j (by omega)).2]

```

Listing 7: Complete proof of Theorem 2, verbatim from final /01-comb08-sidemid-proved.lean (lines 144–159, sha256 19b5c390).

```

1  /-- **T2 (伴随) **: asid 的 mod 2 余数仅依赖 n mod 3, 样式为 pat3
2  (即 asid n 为奇  $\iff n \equiv 0 \pmod{3}$ ; 1-based 口径 = a(n) 为奇  $\iff$ 
3   $n \equiv 1 \pmod{3}$ ) 。 -/
4  theorem asid_mod2_period3 (n r : ℕ) (hr : n % 3 = r) :
5  asid n % 2 = pat3 r := by
6  have main : ∀ n : ℕ, asid n % 2 = pat3 (n % 3) := by
7  intro n
8  induction n using Nat.strong_induction_on with
9  | h n ih =>
10     by_cases hn : n < 6
11     · interval_cases n <=> decide
12     · obtain ⟨j, rfl⟩ : ∃ j, n = j + 6 := ⟨n - 6, by omega⟩
13       have h1 : asid (j + 6) % 2 = asid j % 2 := by
14         simp only [asid]
15         omega
16       rw [h1, ih j (by omega), show (j + 6) % 3 = j % 3 by omega]
17   ]
18   rw [← hr]; exact main n

```

Listing 8: Theorem 3: statement, induction skeleton and the first of the twelve mod_cases branches, verbatim from final /01-comb08-sidemid-proved.lean (lines 163–182, sha256 19b5c390). The remaining eleven branches are identical in shape (lines 183–237) and ship in this deposit under proofs/.

```

1  theorem asid_mod4_period12 (n r : ℕ) (hr : n % 12 = r) :
2  asid n % 4 = pat12 r := by
3  have main : ∀ n : ℕ, asid n % 4 = pat12 (n % 12) := by
4  intro n
5  induction n using Nat.strong_induction_on with
6  | h n ih =>
7     by_cases hn : n < 6
8     · interval_cases n <=> decide
9     · obtain ⟨j, rfl⟩ : ∃ j, n = j + 6 := ⟨n - 6, by omega⟩
10       have h1 : asid (j + 6) % 4
11       = (2 * (asid (j + 4) % 4) + 2 * (asid (j + 2) % 4) +
12       asid j % 4) % 4 := by

```



```

12     simp only [asid]
13     omega
14     rw [h1, ih (j + 4) (by omega), ih (j + 2) (by omega), ih j
15     (by omega)]
16     mod_cases hj : j % 12
17     · have hj0 : j % 12 = 0 := by simp [Nat.ModEq] using hj
18       have h2 : (j + 2) % 12 = 2 := by omega
19       have h4 : (j + 4) % 12 = 4 := by omega
20       have h6 : (j + 6) % 12 = 6 := by omega
       rw [hj0, h2, h4, h6]; decide

```

4 Verification evidence

- Toolchain: Lean v4.34.0 (leanprover/lean4:v4.34.0), mathlib4 v4.34.0, revision 5ed29652 [1, 2, 3]. Reproduction: with that pin, `lake env lean on proofs/01-comb08-sidemid-proved.lean` compiles the file with exit code 0; the strict re-compile of the merged final file measured 29s wall clock in the task ledger.
- Statement integrity: the frozen statement file `formalized/01-comb08-sidemid-statements.lean` (109 lines) has sha256 5fe889d5ed92711a86f23d9827800dd7d2c677a440664f0d81dbdb5e9eb39e67; the final file `final/01-comb08-sidemid-proved.lean` (238 lines) has sha256 19b5c3908592a529491e0b5b40b6b26166019d92820447b41a70ada510a20821. The final file was compared with the frozen file by symmetric difference over the declaration headers: all 8 headers byte-identical, zero drift; the only differences are the three proof bodies replacing the `sorry` placeholders and the two auxiliary lemmas added inside the proof bodies.
- Kernel check: compiles with 0 `sorry` (whole-file scan including comments, case-insensitive, also for `admit` and `sorryAx`); `#print axioms` output, verbatim (kept literal; the smaller monospace width is type-setting only):

```

1 'asid_interleave' depends on axioms: [propext, Quot.sound]
2 'asid_mod2_period3' depends on axioms: [propext, Quot.sound]
3 'asid_mod4_period12' depends on axioms: [propext, Quot.sound]

```

The auxiliary lemmas, probe-verified in the same strict pass, depend on `[propext, Quot.sound]` (`asid_evenlag12`) and `[propext]` (`asid_shift`); `sorryAx` and `Classical.choice` never appear.

- Grading by the audit department: all three theorems *complete proof*; lowest grade reached for the case = *complete proof*; no scaffolding route was used.
- Budgets, copied from the source task ledger’s final machine reconciliation: agent-run sampling 14 of a permitted 20; llm-call 1 of a permitted 6; kernel compiles 26 of a permitted 32; LaTeX rounds 9 of a permitted 10.
- Audit chain (originals in this deposit under `audit/`): final adjudication `final-comb08-sidemid.txt` (four independent checks), the three axiom probe outputs `axioms-asid_interleave.txt`, `axioms-asid_mod2_period3.txt` and `axioms-asid_mod4_period12.txt`, and this note’s listing verification `listing-verify.txt`. The semantic-fidelity check (roundtrip re-translation) of the underlying task ran on the same model node as its formalisation; that independence weakness is mitigated, not removed, by kernel adjudication and manual review.

5 Novelty evidence

Adjudication copied from the pipeline’s exclusivity review (C9 gate), which ran in three rounds around this task and re-derived the data independently rather than trusting the candidate card. Verdict: **no occupying record found** (three rounds consistent); value tier, as corrected by the third round: **new sequence** — the sequence is unoccupied, while its recurrence is *not* new. The full query logs, zero-hit entries included, are in this deposit at `audit/c9-record.md` (verbatim copies of the review records; their raw responses stay in the working repository under `tasks/20260928-comb0709-deeprecheck/`, not shipped in this deposit). Summary by channel:

[illegible]

The highest-risk neighbourhood was closed at the full-text level. Oh’s *State matrix recursion method and monomer–dimer problem* [4] contains, as its Theorem 4, the general formula for coverings with a fixed monomer set — the closest general method in the literature to this note’s object. The statistic is different in shape: Oh’s $g_{m \times n}(S)$ counts coverings whose monomers are *exactly* the fixed set S , whereas $a(n)$ counts *all* matchings of the graph obtained by *deleting* the side-middle cell (any number of monomers). The

full text was read section by section (ar5iv rendering): it contains no instance table, no OEIS reference, and no recurrence for this sequence, and a literal search for twenty-three of the large terms returned a single trivial substring hit. The three nearest references in its bibliography were read in their arXiv versions — Kong [5], Tzeng and Wu [6], Wu [7] — and all three concern the perfect-matching side with a single, unpositioned vacancy, a different statistic again; the twelve citing papers of [4] found in the review contain none on defect-grid full matching counts. The neighbourhood is therefore recorded as a method-level generality, not an occupying record for this sequence.

One correction produced by the third round is part of the record and is load-bearing for this note’s claims: the recurrence has the same characteristic polynomial as the already-registered A033506 [11], signature (4, 14, 0, −10, 0, 1) verbatim, so the earlier rounds’ “new sequence, new recurrence” wording was tightened to “new sequence” — the recurrence’s order and coefficients are not new, only the initial values, that is, the sequence itself. This note’s wording follows the corrected tier, and the correction is the reason no novelty claim is attached to the recurrence anywhere in this note. The two original-language records that fix this wording are quoted verbatim below.

Listing 9: Verbatim quote, original record (Chinese, kept untranslated): the priority fence locked on the source candidate card and repeated in the frozen statement file header (`proofs/01-comb08-sidemid-statements.lean`, lines 18–20).

- 1 占位围栏：价值级仅「新序列」——六阶递推与 A033506（无缺陷 $3 \times n$ 全匹配）同特征多项式（签名 $-(4, 14, 0, 10, 0, 1)$ 逐字相同），禁止对递推阶与系数作新颖性主张； $PM \equiv 0$ 为奇偶+洞色双重否决的真空命题，不构成占位亦不作价值主体。

Listing 10: Verbatim quote, original record (Chinese, kept untranslated): the C9 third-round verdict, from `audit/c9-record.md` (source `deep-recheck-combm-02.md`, section 0 and section 6).

- 1 ****C9 四态：查无占位（三轮一致，本审支持前两审结论）。****
- 2 ****价值分级：新序列——「新递推」主张经本审机器判定不成立（递推与已挂名的 A033506 同特征多项式），这是对前两审分级措辞的修正（收紧，不影响四态裁决）。****
- 3 - ****C9 四态：查无占位**（证据链同共享记录 §8）。**
- 4 - ****价值分级：新序列**（递推非新——与 A033506 同谱；「新序列·新递推」措辞按本审修正）。派生 $PM \equiv 0$ 为经典着色阻塞实例，不构成占位亦不作价值主体。**

The perfect-matching side of the neighbourhood is already occupied for neighbouring defect configurations: the even- n perfect-matching subsequence of one neighbouring defect is A061278 [14], and the perfect-matching

subsequence of another neighbouring defect configuration is A129113 [15]. For this note’s configuration the perfect-matching count is identically zero, so there is no subsequence to occupy and nothing to claim; the two registry entries are reported as context, not as prior art for any statement of this note.

Given that evidence this note claims what the review supports: the sequence was not found to be occupied. The novelty of Theorems 1, 2 and 3 is consequently negative-search evidence, not a first-discovery assertion, and should be read as such. Disclosed holes: the general web-search layer was unreachable in the review environment, so the literature evidence rests on OpenAlex, zbMATH, arXiv, Crossref and engine-side queries; the mother-family monographs [9] were identified through the registry records but their full texts were not read; one paywalled reference was not read; the Math Stack Exchange native API was throttled; and sequencelib is structurally absent for sequences without an A-number. If those channels later return a hit on this sequence, the positioning in this section weakens accordingly; the theorems themselves do not depend on it.

6 Scope and limitations

Grade and boundary, stated as the audit states them. (a) The theorems cover the recurrence-defined sequences only; their equality with the tiling counts of the side-defected grid is not kernel-proved and is Conjecture 1, whose empirical support is enumeration by exact programs (three structurally different methods; seeds $n \leq 14$ reproduced digit-for-digit; deep window $n \leq 50$; the enumerators themselves are not verified, and the scripts and their outputs stay in the working repository under `tasks/20260928-comb08-sidemid/`). (b) The recurrence’s order and coefficients are not new: its characteristic polynomial is shared with the registered A033506 [11], signature $(4, 14, 0, -10, 0, 1)$ verbatim, and no novelty is claimed for them anywhere in this note; what is claimed is the sequence itself, at the value tier *new sequence*. (c) The perfect-matching count of this configuration is identically zero — a vacuum proposition forced by parity and by the classical colouring obstruction — and is reported as neither an occupying record nor a value subject. (d) The gate-four sign-off of the underlying task was exercised by the author’s instruction to deposit this claim on 2026-09-28, while the task report file still carries its draft header reading, in the original, “awaiting manual sign-off at gate four”; both facts are recorded here as they stand, and the `tasks/` products were

not modified. (e) This note carries no literature review and is a deposit record, not a survey; the narrative paper for the same result is separate.

Data availability

The deposit package for this claim contains the claim note (LaTeX source and PDF), the Lean sources (`proofs/`: frozen statement file, final proof file, `lean-toolchain`), the verification and audit logs, and the full C9 query log with zero-hit entries (`audit/`). Zenodo: the DOI is reserved at upload by the author; that field is currently unfilled and is written `RESERVED-DOI` in this note, to be replaced by the reserved DOI in this note, in `metadata/README.md` of this package and in a recompiled PDF at upload time; the manual steps are listed in `UPLOAD.md` in the source repository (not shipped in this deposit). Public mirror of this deposit: the claim lane’s repository <https://github.com/Aurora0134/ai4math-claims>, directory `grid3n-sidemid-notch/` (pushed under the author’s instruction of 2026-09-28; the commit and timestamp are recorded in the source repository’s `claims/grid3n-sidemid-notch/card.md`). Claim note under CC BY 4.0, code under MIT.

Use of generative AI

(a) AI performed: candidate generation and the exclusivity review, autoformalisation of the three statements, the roundtrip semantic check, proof search and proof-script writing, error classification and repair, and the assembly of this note including its verbatim listings. (b) Model, node and budget: all task-stage sampling ran on one node, wire-id `anthropic/a6api-main/kimi-k3` (the task card’s snapshot; a same-day drift observation to `anthropic/stepfun/step-5-preview` is disclosed in the source task’s ledger rather than replaced), with no cross-node switching, and consumed 14 agent-run samplings of a permitted 20, 1 `llm-call` of a permitted 6, 26 kernel compiles of a permitted 32 and 9 LaTeX rounds of a permitted 10, copied verbatim from the task ledger’s final reconciliation. This note’s assembly consumed 0 pipeline LLM calls against a cap of 4 and was dispatched as one agent-run sampling on the current session node (`stepfun/step-5-preview`). (c) Final adjudication: all statements and proofs reported here were checked by the Lean 4 kernel: every proof compiles with no `sorry`, and `#print axioms` reports only `{propext, Quot.sound}`, a subset of the standard whitelist `{propext, Quot.sound,`

Classical.choice}. (d) The AI system is not an author; the human author reviewed the evidence and is responsible for all content.

Author contributions

CRediT-style: the human author adjudicated topic selection, statement semantics and the deposit decision; the AI4Math pipeline (an LLM) generated the candidate propositions, the proof searches and the text drafts. The AI system is not listed as an author.

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- [11] The OEIS Foundation. Entry A033506 (P. H. Lundow): number of matchings in graph $P_3 \times P_n$; index entry for linear recurrences with constant coefficients, signature (4, 14, 0, −10, 0, 1). *The On-Line Encyclopedia of Integer Sequences*. <https://oeis.org/A033506>
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